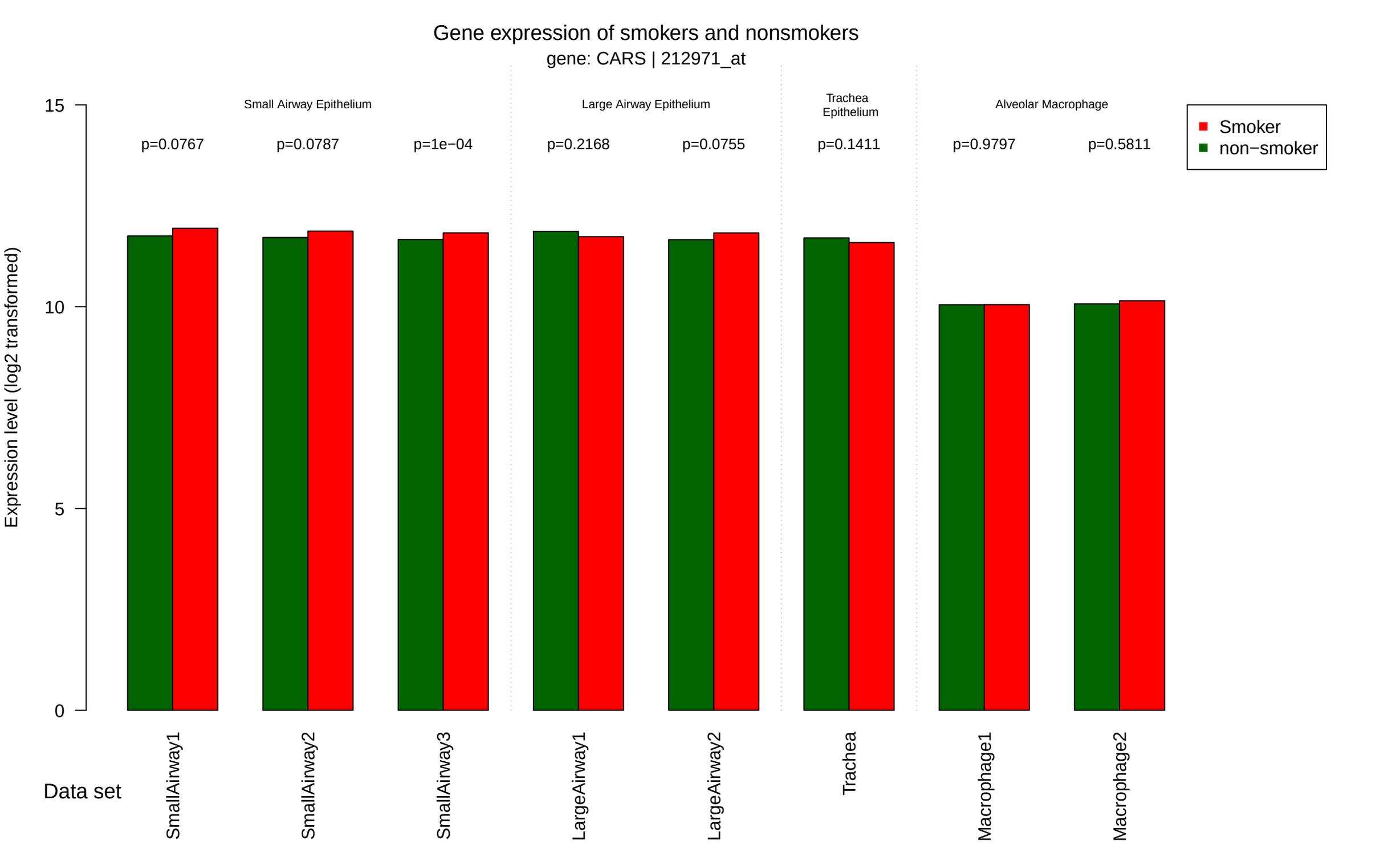
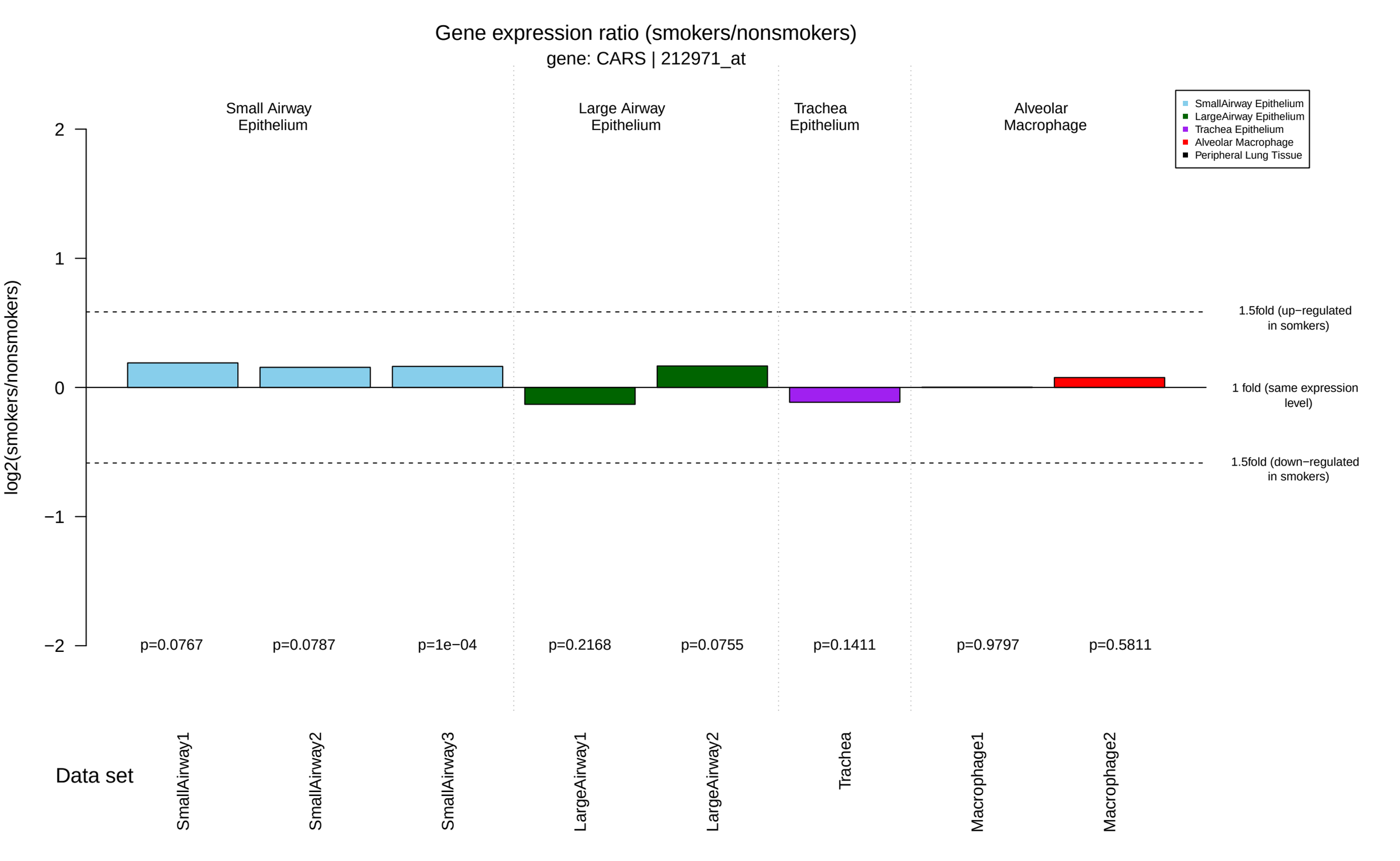
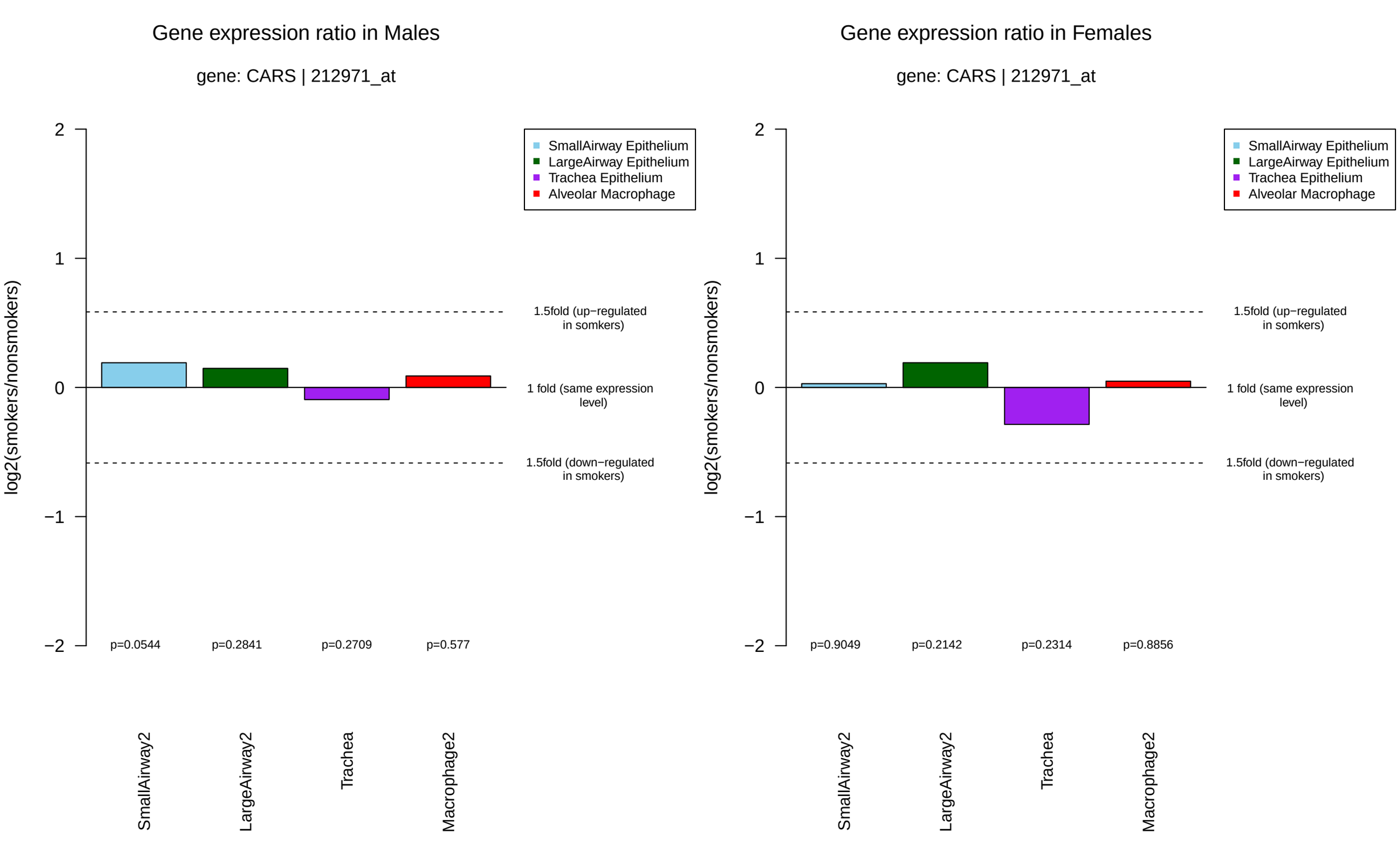
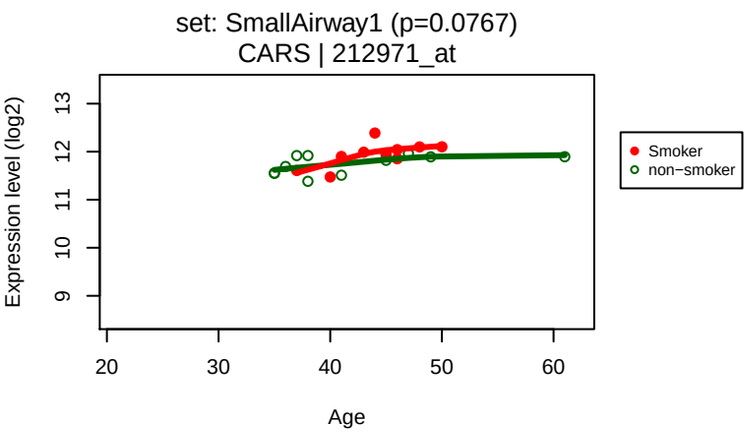
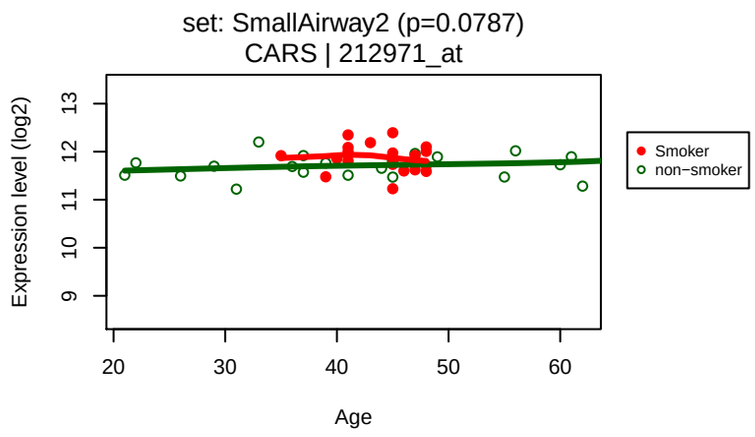
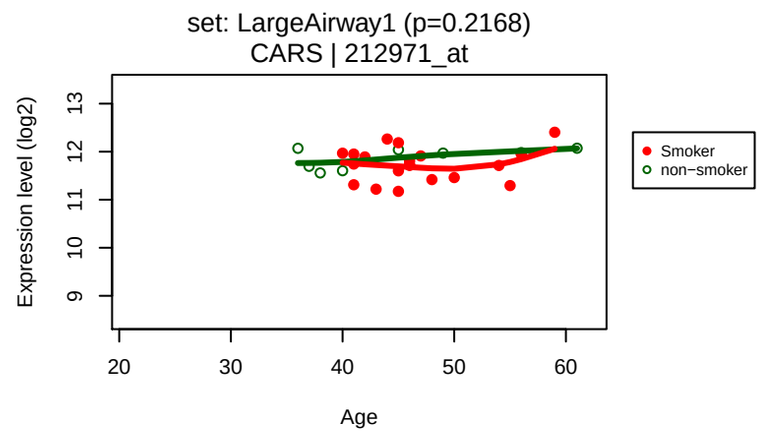

Gene expression of smokers and nonsmokers
gene: CARS | 212971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0767
p=0.0787
p=1e−04
p=0.2168
p=0.0755
p=0.1411
p=0.9797
p=0.5811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

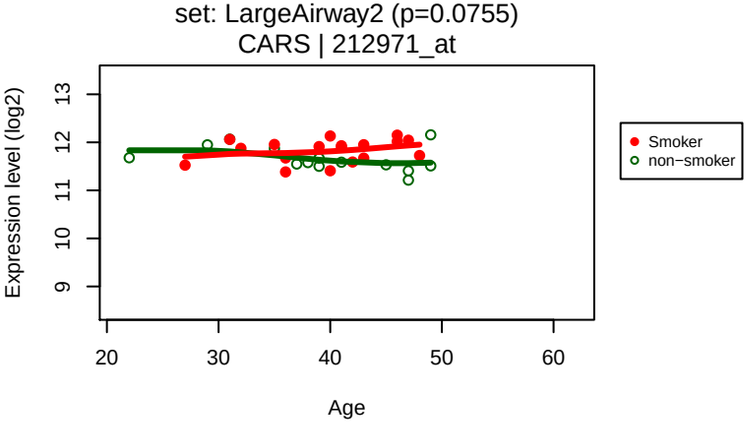

Gene expression ratio (smokers/nonsmokers)
gene: CARS | 212971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0767
p=0.0787
p=1e−04
p=0.2168
p=0.0755
p=0.1411
p=0.9797
p=0.5811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

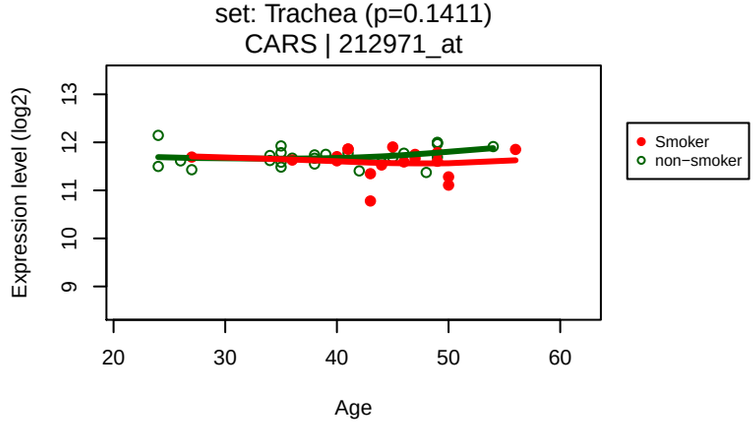

Gene expression ratio in Males
gene: CARS | 212971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0544
p=0.2841
p=0.2709
p=0.577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS | 212971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9049
p=0.2142
p=0.2314
p=0.8856
SmallAirway2
LargeAirway2
Trachea
Macrophage2

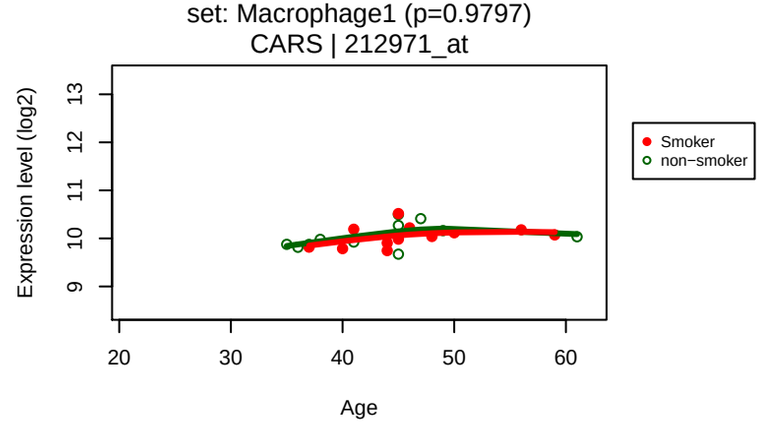

set: SmallAirway1 (p=0.0767)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker

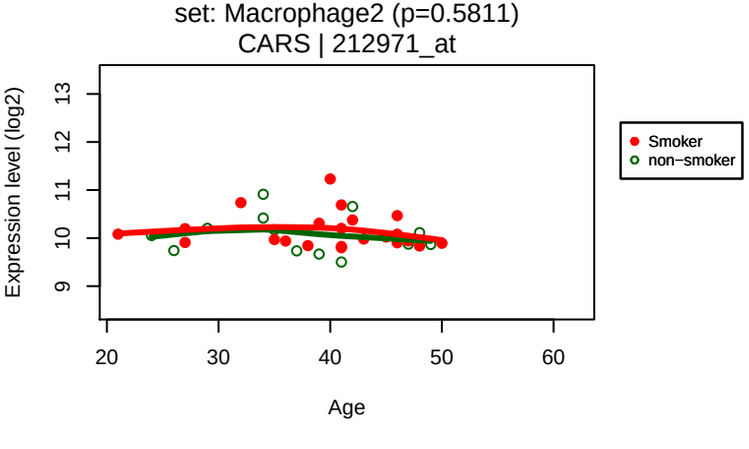

set: SmallAirway2 (p=0.0787)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker

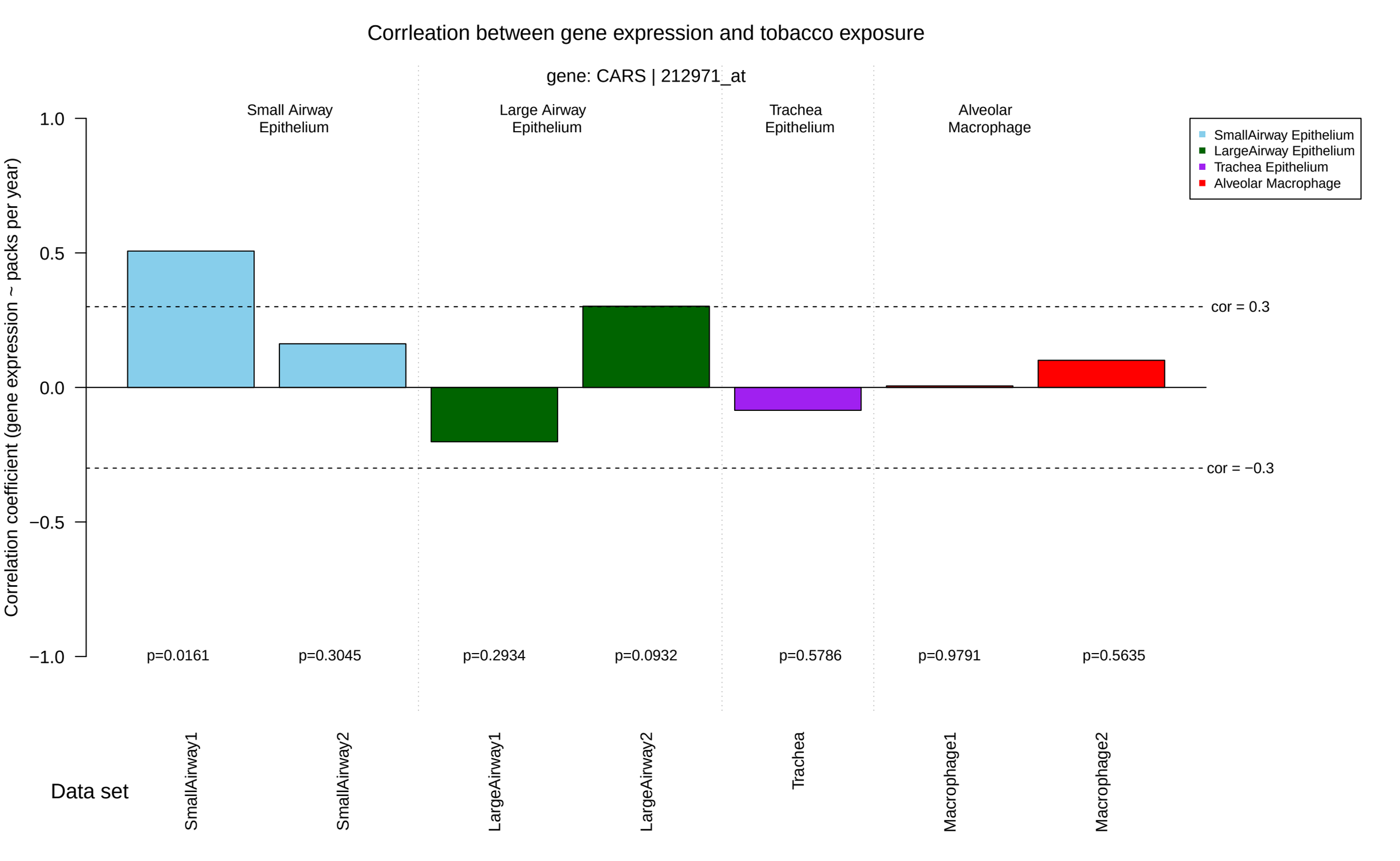

set: LargeAirway1 (p=0.2168)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0755)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1411)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9797)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5811)
CARS | 212971_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS | 212971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0161
p=0.3045
p=0.2934
p=0.0932
p=0.5786
p=0.9791
p=0.5635
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2